

## 1.10 Kronecker products and the vec operator

Balanced designs have model matrices with a product structure, and multivariate observations have covariance matrices with one. The Kronecker product expresses both.

### Definition 1.10.1

The **Kronecker product** of an  $m \times n$  matrix  $\mathbf{A}$  and a  $p \times q$  matrix  $\mathbf{B}$  is the  $mp \times nq$  block matrix  $\mathbf{A} \otimes \mathbf{B} = (a_{ij}\mathbf{B})$ . The **vec operator** places the  $n$  columns of an  $m \times n$  matrix one below another, giving a vector of length  $mn$ :  $\text{vec}(\mathbf{A}) = (\mathbf{a}_1^\top, \dots, \mathbf{a}_n^\top)^\top$ .

### Proposition 1.10.1

For matrices of compatible sizes:

- (a)  $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = \mathbf{AC} \otimes \mathbf{BD}$  (mixed-product rule);
- (b)  $(\mathbf{A} \otimes \mathbf{B})^\top = \mathbf{A}^\top \otimes \mathbf{B}^\top$ , and  $(\mathbf{A} \otimes \mathbf{B})^{-1} = \mathbf{A}^{-1} \otimes \mathbf{B}^{-1}$  for nonsingular  $\mathbf{A}, \mathbf{B}$ . More generally  $\mathbf{A}^- \otimes \mathbf{B}^-$  is a generalized inverse of  $\mathbf{A} \otimes \mathbf{B}$ ;
- (c)  $\text{tr}(\mathbf{A} \otimes \mathbf{B}) = \text{tr}(\mathbf{A}) \text{tr}(\mathbf{B})$  and  $\text{rank}(\mathbf{A} \otimes \mathbf{B}) = \text{rank}(\mathbf{A}) \text{rank}(\mathbf{B})$ ;
- (d) if  $\mathbf{A}$  ( $m \times m$ ) and  $\mathbf{B}$  ( $p \times p$ ) are symmetric with eigenvalues  $\lambda_i$  and  $\mu_j$ , the eigenvalues of  $\mathbf{A} \otimes \mathbf{B}$  are the products  $\lambda_i \mu_j$ , and  $\det(\mathbf{A} \otimes \mathbf{B}) = (\det \mathbf{A})^p (\det \mathbf{B})^m$ ;
- (e) Kronecker products of orthogonal, symmetric, idempotent, or nonnegative definite matrices are of the same kind;
- (f)  $\text{vec}(\mathbf{ABC}) = (\mathbf{C}^\top \otimes \mathbf{A}) \text{vec}(\mathbf{B})$  and  $\text{tr}(\mathbf{A}^\top \mathbf{B}) = \text{vec}(\mathbf{A})^\top \text{vec}(\mathbf{B})$ .

*Proof.*

- (a) The  $(i, j)$  block of the left side is  $\sum_k a_{ik} c_{kj} \mathbf{B}$ , which is the  $(i, j)$  block of the right side.
- (b) Transposition is direct, and the rest follows from (a), for example  $(\mathbf{A} \otimes \mathbf{B})(\mathbf{A}^- \otimes \mathbf{B}^-)(\mathbf{A} \otimes \mathbf{B}) = \mathbf{AA}^- \otimes \mathbf{BB}^- \mathbf{B}$ .
- (b) With  $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^\top$  and  $\mathbf{B} = \mathbf{Q}\mathbf{M}\mathbf{Q}^\top$ , (a) and (b) give  $\mathbf{A} \otimes \mathbf{B} = (\mathbf{P} \otimes \mathbf{Q})(\mathbf{\Lambda} \otimes \mathbf{M})(\mathbf{P} \otimes \mathbf{Q})^\top$ , a spectral decomposition because  $\mathbf{P} \otimes \mathbf{Q}$  is orthogonal and  $\mathbf{\Lambda} \otimes \mathbf{M}$  is diagonal with entries  $\lambda_i \mu_j$ . The determinant is the product of these. (c) The trace statement is direct. For the rank, by (a)  $(\mathbf{A} \otimes \mathbf{B})(\mathbf{A}^- \otimes \mathbf{B}^-) = \mathbf{AA}^- \otimes \mathbf{BB}^-$  is idempotent with the same rank as  $\mathbf{A} \otimes \mathbf{B}$  (Proposition 1.3.3(a)). Its trace is  $\text{tr}(\mathbf{AA}^-) \text{tr}(\mathbf{BB}^-) = \text{rank}(\mathbf{A}) \text{rank}(\mathbf{B})$  by Proposition 1.9.1(b).
- (c) follows from (a), (b) and (d). (f) The  $j$ th column of  $\mathbf{ABC}$  is  $\mathbf{ABc}_j = \sum_k c_{kj} \mathbf{Ab}_k$ , which is the  $j$ th block of  $(\mathbf{C}^\top \otimes \mathbf{A}) \text{vec}(\mathbf{B})$ . The trace identity is  $\sum_{ij} a_{ij} b_{ij}$  written two ways.

■

The determinant formula in (d) holds for all square  $\mathbf{A}$  and  $\mathbf{B}$ , not only symmetric ones. We need only the symmetric case.

**Example. The balanced two-way layout**

Let observations  $y_{ij}$ ,  $i = 1, \dots, a$ ,  $j = 1, \dots, b$ , be stacked with  $j$  running inside  $i$ . The additive model  $E(y_{ij}) = \mu + \alpha_i + \beta_j$  has model matrix

$$\mathbf{X} = [\mathbf{1}_a \otimes \mathbf{1}_b, \mathbf{I}_a \otimes \mathbf{1}_b, \mathbf{1}_a \otimes \mathbf{I}_b],$$

because  $\mathbf{I}_a \otimes \mathbf{1}_b$  has a 1 in column  $i$  exactly for the  $b$  observations in row  $i$ . With  $\bar{\mathbf{J}}_k = k^{-1} \mathbf{J}_k$ , a symmetric idempotent averaging matrix, the mixed-product rule shows that  $\mathbf{I}_a \otimes \bar{\mathbf{J}}_b$ ,  $\bar{\mathbf{J}}_a \otimes \mathbf{I}_b$  and  $\bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b$  replace each observation by its row mean, its column mean and the grand mean. The matrix  $\mathbf{M} = \mathbf{I}_a \otimes \bar{\mathbf{J}}_b + \bar{\mathbf{J}}_a \otimes \mathbf{I}_b - \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b$  is symmetric and idempotent, with trace  $a + b - 1 = \text{rank}(\mathbf{X})$ . It is the projection onto  $\mathcal{C}(\mathbf{X})$  (Theorem 16.1.2 in Chapter 16). The script checks this for  $a = 3$  and  $b = 4$ , together with each part of Proposition 1.10.1.

```
import numpy as np
rng = np.random.default_rng(11)
def vec(M):
    return M.reshape(-1, order="F")          # stack the columns

A, B, C = rng.normal(size=(3, 4)), rng.normal(size=(4, 2)), rng.normal(size=(2, 5))
assert np.allclose(vec(A @ B @ C), np.kron(C.T, A) @ vec(B))          # vec(ABC)

# balanced two-way layout: a = 3 rows, b = 4 columns, one observation per cell
a, b = 3, 4
one = lambda k: np.ones((k, 1))
X = np.hstack([np.kron(one(a), one(b)),      # intercept
               np.kron(np.eye(a), one(b)),  # row indicators
               np.kron(one(a), np.eye(b))])  # column indicators
print("rank(X) =", np.linalg.matrix_rank(X), "= a + b - 1 =", a + b - 1)
```

[Run this code online](#)

## 1.10.1 Exercises

### 1.10.1.1 A. Check your understanding

#### Exercise. A1

With observations stacked as in Example (The balanced two-way layout), the row and column indicator matrices are  $\mathbf{R} = \mathbf{I}_a \otimes \mathbf{1}_b$  and  $\mathbf{C} = \mathbf{1}_a \otimes \mathbf{I}_b$ . Use the mixed-product rule to compute  $\mathbf{R}^\top \mathbf{R}$ ,  $\mathbf{C}^\top \mathbf{C}$  and  $\mathbf{R}^\top \mathbf{C}$ , and identify the cross-product matrix  $\mathbf{X}^\top \mathbf{X}$  of the balanced additive model.

*Solution.* By Proposition 1.10.1(a) and (b),  $\mathbf{R}^\top \mathbf{R} = (\mathbf{I}_a^\top \mathbf{I}_a) \otimes (\mathbf{1}_b^\top \mathbf{1}_b) = b \mathbf{I}_a$  and  $\mathbf{C}^\top \mathbf{C} = (\mathbf{1}_a^\top \mathbf{1}_a) \otimes \mathbf{I}_b = a \mathbf{I}_b$ , while  $\mathbf{R}^\top \mathbf{C} = (\mathbf{I}_a^\top \mathbf{1}_a) \otimes (\mathbf{1}_b^\top \mathbf{I}_b) = \mathbf{1}_a \mathbf{1}_b^\top$ , the  $a \times b$  matrix of ones. With  $\mathbf{1}_{ab} = \mathbf{1}_a \otimes \mathbf{1}_b = \mathbf{R} \mathbf{1}_a = \mathbf{C} \mathbf{1}_b$ , the cross-product matrix of  $\mathbf{X} = [\mathbf{1}_{ab}, \mathbf{R}, \mathbf{C}]$  is

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} ab & b \mathbf{1}_a^\top & a \mathbf{1}_b^\top \\ b \mathbf{1}_a & b \mathbf{I}_a & \mathbf{1}_a \mathbf{1}_b^\top \\ a \mathbf{1}_b & \mathbf{1}_b \mathbf{1}_a^\top & a \mathbf{I}_b \end{pmatrix}.$$

Every cell of the layout is filled once, so each pair of a row level and a column level occurs

exactly once, which is what the block of ones records.

### Exercise. A2

Show that  $(\mathbf{A} \otimes \mathbf{B})^\top (\mathbf{A} \otimes \mathbf{B}) = (\mathbf{A}^\top \mathbf{A}) \otimes (\mathbf{B}^\top \mathbf{B})$ , and deduce that the Kronecker product of two matrices with orthonormal columns again has orthonormal columns. Show also that  $\|\mathbf{A}\|_F = \|\text{vec}(\mathbf{A})\|$ , so that  $\text{vec}$  turns the Frobenius norm into the Euclidean norm.

### Exercise. A3

Let  $\Sigma = \Sigma_1 \otimes \Sigma_2$  with  $\Sigma_1$  ( $m \times m$ ) and  $\Sigma_2$  ( $p \times p$ ) positive definite — a *separable* covariance matrix, one variance structure across rows and another across columns. Show that  $\Sigma$  is positive definite, that  $\Sigma^{-1} = \Sigma_1^{-1} \otimes \Sigma_2^{-1}$ , that  $\Sigma^{1/2} = \Sigma_1^{1/2} \otimes \Sigma_2^{1/2}$  in the sense of Theorem 1.7.6, and that  $\log \det \Sigma = p \log \det \Sigma_1 + m \log \det \Sigma_2$ .

## 1.10.1.2 B. Practice

### Exercise. B1

Let  $\mathbf{Y}$  be  $n \times q$ , let  $\mathbf{X}$  be  $n \times p$  with full column rank and let  $\mathbf{B}$  be  $p \times q$ , so that  $q$  responses are regressed on the same  $\mathbf{X}$ . Using Proposition 1.10.1(f), show that  $\text{vec}(\mathbf{XB}) = (\mathbf{I}_q \otimes \mathbf{X}) \text{vec}(\mathbf{B})$  and that minimizing  $\|\mathbf{Y} - \mathbf{XB}\|_F^2$  leads to

$$(\mathbf{I}_q \otimes \mathbf{X}^\top \mathbf{X}) \text{vec}(\mathbf{B}) = \text{vec}(\mathbf{X}^\top \mathbf{Y}).$$

Deduce that a multivariate regression on a common model matrix is  $q$  separate univariate regressions, one per column of  $\mathbf{Y}$ .

*Solution.* Part (f) with  $\mathbf{XB}\mathbf{I}_q$  gives  $\text{vec}(\mathbf{XB}) = (\mathbf{I}_q^\top \otimes \mathbf{X}) \text{vec}(\mathbf{B})$ . By Exercise (A2),  $\|\mathbf{Y} - \mathbf{XB}\|_F^2 = \|\text{vec}(\mathbf{Y}) - (\mathbf{I}_q \otimes \mathbf{X}) \text{vec}(\mathbf{B})\|^2$ , an ordinary least squares criterion with model matrix  $\mathbf{I}_q \otimes \mathbf{X}$ . Its normal equations are

$$(\mathbf{I}_q \otimes \mathbf{X})^\top (\mathbf{I}_q \otimes \mathbf{X}) \text{vec}(\mathbf{B}) = (\mathbf{I}_q \otimes \mathbf{X})^\top \text{vec}(\mathbf{Y}),$$

and the mixed-product rule turns the left side into  $(\mathbf{I}_q \otimes \mathbf{X}^\top \mathbf{X}) \text{vec}(\mathbf{B})$  and the right side into  $\text{vec}(\mathbf{X}^\top \mathbf{Y})$ . The matrix  $\mathbf{I}_q \otimes \mathbf{X}^\top \mathbf{X}$  is block diagonal with identical blocks, so the system separates into  $\mathbf{X}^\top \mathbf{X} \mathbf{b}_j = \mathbf{X}^\top \mathbf{y}_j$  for each column  $\mathbf{y}_j$  of  $\mathbf{Y}$ . Nothing is gained by treating the responses jointly, as long as they share a model matrix and the criterion is the unweighted sum of squares.

### Exercise. B2

Show that  $\mathbf{A} \otimes \mathbf{B} = \mathbf{0}$  iff  $\mathbf{A} = \mathbf{0}$  or  $\mathbf{B} = \mathbf{0}$ , and that if  $\mathbf{A} \otimes \mathbf{B} = \mathbf{C} \otimes \mathbf{D}$  with  $\mathbf{A}, \mathbf{C}$  of one size and  $\mathbf{B}, \mathbf{D}$  of another, all four nonzero, then  $\mathbf{C} = c^{-1}\mathbf{A}$  and  $\mathbf{D} = c\mathbf{B}$  for some scalar  $c \neq 0$ . A separable covariance matrix therefore determines its two factors only up to reciprocal scalars.

*Solution.* The  $(i, j)$  block of  $\mathbf{A} \otimes \mathbf{B}$  is  $a_{ij}\mathbf{B}$ , so the product is zero iff every  $a_{ij}\mathbf{B} = \mathbf{0}$ . If  $\mathbf{A} \otimes \mathbf{B} = \mathbf{C} \otimes \mathbf{D}$ , then  $a_{ij}\mathbf{B} = c_{ij}\mathbf{D}$  for all  $i, j$ . Choose  $k, l$  with  $a_{kl} \neq 0$ . If  $c_{kl} = 0$  then  $\mathbf{B} = \mathbf{0}$ , which is excluded, so  $\mathbf{D} = c\mathbf{B}$  with  $c = a_{kl}/c_{kl} \neq 0$ . Substituting back,  $a_{ij}\mathbf{B} = c c_{ij}\mathbf{B}$  for all  $i, j$ , and  $\mathbf{B} \neq \mathbf{0}$  forces  $a_{ij} = c c_{ij}$ , that is,  $\mathbf{C} = c^{-1}\mathbf{A}$ . Conversely  $(c^{-1}\mathbf{A}) \otimes (c\mathbf{B}) = \mathbf{A} \otimes \mathbf{B}$ .

### Exercise. B3

Write  $\bar{\mathbf{J}}_k = k^{-1}\mathbf{J}_k$  and  $\mathbf{Q}_k = \mathbf{I}_k - \bar{\mathbf{J}}_k$ , both symmetric idempotent, of ranks 1 and  $k - 1$ . Expanding  $\mathbf{I}_{ab} = (\bar{\mathbf{J}}_a + \mathbf{Q}_a) \otimes (\bar{\mathbf{J}}_b + \mathbf{Q}_b)$ , show that the four matrices

$$\bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b, \quad \mathbf{Q}_a \otimes \bar{\mathbf{J}}_b, \quad \bar{\mathbf{J}}_a \otimes \mathbf{Q}_b, \quad \mathbf{Q}_a \otimes \mathbf{Q}_b$$

are symmetric idempotent, sum to  $\mathbf{I}_{ab}$  and multiply to zero in pairs, with ranks 1,  $a - 1$ ,  $b - 1$  and  $(a - 1)(b - 1)$ . Show that the matrix  $\mathbf{M}$  of [Example \(The balanced two-way layout\)](#) is the sum of the first three, so that  $\mathbf{I} - \mathbf{M} = \mathbf{Q}_a \otimes \mathbf{Q}_b$ . These are the grand mean, row, column and interaction parts of a two-way analysis of variance, and their ranks are its degrees of freedom.

*Solution.* Each of the four is a Kronecker product of symmetric idempotent matrices, hence symmetric idempotent by [Proposition 1.10.1\(e\)](#), with rank the product of the ranks by [Proposition 1.10.1\(c\)](#). Expanding the product of the two decompositions of the identity gives  $\mathbf{I}_a \otimes \mathbf{I}_b$  as their sum. Any two of them have a factor  $\bar{\mathbf{J}}\mathbf{Q} = \mathbf{0}$  or  $\mathbf{Q}\bar{\mathbf{J}} = \mathbf{0}$  in one of the two slots, so their product vanishes by the mixed-product rule. The four ranks add to  $1 + (a - 1) + (b - 1) + (a - 1)(b - 1) = ab$ , as Theorem 1.9.3 requires. Finally

$$\mathbf{M} = \mathbf{I}_a \otimes \bar{\mathbf{J}}_b + \bar{\mathbf{J}}_a \otimes \mathbf{I}_b - \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b = (\bar{\mathbf{J}}_a + \mathbf{Q}_a) \otimes \bar{\mathbf{J}}_b + \bar{\mathbf{J}}_a \otimes (\bar{\mathbf{J}}_b + \mathbf{Q}_b) - \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b,$$

which collects into  $\bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b + \mathbf{Q}_a \otimes \bar{\mathbf{J}}_b + \bar{\mathbf{J}}_a \otimes \mathbf{Q}_b$ , of rank  $1 + (a - 1) + (b - 1) = a + b - 1$ . Subtracting from  $\mathbf{I}_{ab}$  leaves  $\mathbf{Q}_a \otimes \mathbf{Q}_b$ .

### 1.10.1.3 C. Going deeper

#### Exercise. C1

Show that for each  $m, n$  there is exactly one  $mn \times mn$  matrix  $\mathbf{K}_{m,n}$  with  $\mathbf{K}_{m,n} \text{vec}(\mathbf{A}) = \text{vec}(\mathbf{A}^\top)$  for every  $m \times n$  matrix  $\mathbf{A}$ , that it is a permutation matrix, and that  $\mathbf{K}_{m,n}^\top = \mathbf{K}_{m,n}^{-1} = \mathbf{K}_{n,m}$ . Then show that for  $\mathbf{A}$  of size  $m \times n$  and  $\mathbf{B}$  of size  $p \times q$ ,

$$\mathbf{K}_{p,m}(\mathbf{A} \otimes \mathbf{B}) = (\mathbf{B} \otimes \mathbf{A})\mathbf{K}_{q,n}.$$

This **commutation matrix** is how the two orders of a Kronecker product are related, and how derivatives with respect to a matrix are rearranged.

*Solution.* The map  $\text{vec}(\mathbf{A}) \mapsto \text{vec}(\mathbf{A}^\top)$  permutes the  $mn$  coordinates: the entry  $a_{ij}$ , which sits in position  $i + (j - 1)m$  of  $\text{vec}(\mathbf{A})$ , sits in position  $j + (i - 1)n$  of  $\text{vec}(\mathbf{A}^\top)$ . A linear map on  $\mathbb{R}^{mn}$  is determined by its values on the coordinate vectors, and every coordinate vector is  $\text{vec}(\mathbf{e}_i \mathbf{e}_j^\top)$  for some  $i, j$ , so  $\mathbf{K}_{m,n}$  exists and is unique, and it is the permutation matrix just described. Applying the rule twice returns  $\text{vec}(\mathbf{A})$ , so  $\mathbf{K}_{n,m}\mathbf{K}_{m,n} = \mathbf{I}$ ; a permutation matrix

is orthogonal, so its inverse is its transpose.

For the last identity, let  $\mathbf{C}$  be  $q \times n$ , so that both sides may be applied to  $\text{vec}(\mathbf{C})$ . By [Proposition 1.10.1\(f\)](#),  $(\mathbf{A} \otimes \mathbf{B}) \text{vec}(\mathbf{C}) = \text{vec}(\mathbf{BCA}^\top)$ , a  $p \times m$  matrix, so the left side sends  $\text{vec}(\mathbf{C})$  to  $\text{vec}((\mathbf{BCA}^\top)^\top) = \text{vec}(\mathbf{AC}^\top \mathbf{B}^\top)$ . The right side sends it to  $(\mathbf{B} \otimes \mathbf{A}) \text{vec}(\mathbf{C}^\top) = \text{vec}(\mathbf{AC}^\top \mathbf{B}^\top)$  by the same rule. The two matrices agree on every  $\text{vec}(\mathbf{C})$ , that is, on all of  $\mathbb{R}^{nq}$ , so they are equal.

### Exercise. C2

In a balanced two-way layout with random row and column effects, the response vector has covariance

$$\mathbf{V} = \sigma_\alpha^2(\mathbf{I}_a \otimes \mathbf{J}_b) + \sigma_\beta^2(\mathbf{J}_a \otimes \mathbf{I}_b) + \sigma^2 \mathbf{I}_{ab}, \quad \sigma^2 > 0.$$

Using the four idempotent matrices of [Exercise \(B3\)](#), show that

$$\mathbf{V} = (b\sigma_\alpha^2 + a\sigma_\beta^2 + \sigma^2) \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b + (b\sigma_\alpha^2 + \sigma^2) \mathbf{Q}_a \otimes \bar{\mathbf{J}}_b + (a\sigma_\beta^2 + \sigma^2) \bar{\mathbf{J}}_a \otimes \mathbf{Q}_b + \sigma^2 \mathbf{Q}_a \otimes \mathbf{Q}_b,$$

read off the eigenvalues of  $\mathbf{V}$  with their multiplicities, and write down  $\det \mathbf{V}$  and  $\mathbf{V}^{-1}$ . On which subspace does  $\mathbf{V}$  act as  $\sigma^2 \mathbf{I}$ ?

*Solution.* Since  $\mathbf{J}_k = k\bar{\mathbf{J}}_k$  and  $\mathbf{I}_k = \bar{\mathbf{J}}_k + \mathbf{Q}_k$ ,

$$\mathbf{I}_a \otimes \mathbf{J}_b = b(\bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b + \mathbf{Q}_a \otimes \bar{\mathbf{J}}_b), \quad \mathbf{J}_a \otimes \mathbf{I}_b = a(\bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b + \bar{\mathbf{J}}_a \otimes \mathbf{Q}_b),$$

and  $\mathbf{I}_{ab}$  is the sum of all four. Collecting coefficients gives the displayed form. The four matrices are symmetric idempotent with pairwise products zero and sum  $\mathbf{I}_{ab}$ , so by Theorem 1.9.2(a) the columns of an orthonormal basis of each column space are eigenvectors of  $\mathbf{V}$ , with eigenvalue the coefficient in front. The eigenvalues are therefore  $b\sigma_\alpha^2 + a\sigma_\beta^2 + \sigma^2$  once,  $b\sigma_\alpha^2 + \sigma^2$  with multiplicity  $a - 1$ ,  $a\sigma_\beta^2 + \sigma^2$  with multiplicity  $b - 1$  and  $\sigma^2$  with multiplicity  $(a - 1)(b - 1)$ . The determinant is their product,

$$\det \mathbf{V} = (b\sigma_\alpha^2 + a\sigma_\beta^2 + \sigma^2)(b\sigma_\alpha^2 + \sigma^2)^{a-1}(a\sigma_\beta^2 + \sigma^2)^{b-1}(\sigma^2)^{(a-1)(b-1)},$$

and the inverse replaces each coefficient in the displayed sum by its reciprocal, since the four terms multiply to zero in pairs and each is idempotent. Finally  $\mathbf{V}\mathbf{u} = \sigma^2 \mathbf{u}$  for every  $\mathbf{u} \in \mathcal{C}(\mathbf{Q}_a \otimes \mathbf{Q}_b)$ : on the interaction space the row and column variance components have no effect at all, which is why that space supplies the error term of the balanced two-way analysis of variance.